
HELP THE LOW SKILLED OR LET THE HARDWORKING THRIVE? A STUDY OF FAIRNESS IN OPTIMAL INCOME TAXATION

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Abstract

In a model where agents have unequal wages and heterogeneous preferences, we study the optimal redistribution via an income tax, when the social objective is based on a combination of efficiency and fairness principles, and when incentive issues are taken into account. We show how some fairness principles entail specific features for the optimal taxes, such as progressivity or tax exemption for incomes below the minimum wage.

1. Introduction

Redistribution via an income tax does not only involve considerations of inequality reduction or of efficiency losses but also and, maybe primarily, of fairness. Is it fair to try to reduce income inequalities indiscriminately, when some are due to unequal earning abilities while others are simply due to

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different consumption-leisure preferences? Is it fair to impose a high marginal tax rate on those who have the lowest wages? In this paper, we analyze these issues by establishing a rigorous connection between axioms of fairness and features of the income tax schedule.

We consider a model in which individuals differ in their earning ability as well as their consumption-leisure preferences. We show how to define social preferences that incorporate efficiency concerns and fairness requirements. Two main fairness requirements are referred to here. The first one defends inequality reductions between agents who have identical preferences and differ in their earning ability. The second one is oriented toward the respect of individual consumption-leisure preferences and goes against redistribution between agents with the same earning ability. These two requirements can be related to ethical principles that are well known in the fairness literature, and it has now been well established that there is a tension between them (see, e.g., Fleurbaey 1995). But this tension offers an interesting ethical choice about which value should have priority over the other: Help the low skilled agents, or let the hardworking reap the benefits of their work? This dilemma has echoes in recent debates about the welfare state. Depending on which value is given the lead, one obtains two different kinds of social objectives, leading potentially to different tax schedules.

In this paper, we study the consequences of such alternative social preferences for tax design. First we briefly summarize the axiomatic analysis of fairness requirements and of the social preferences which can be characterized on their basis (see Fleurbaey and Maniquet 2005 for the detailed study), and then we focus on the derivation of the optimal tax.

The computation of the optimal tax for a population which is heterogeneous in two dimensions (skill and preferences) is a difficult exercise for which no general solution exists.¹ But we will be able to describe at least what part of the agents' budget set created by the tax schedule should be the focus of the social planner and to determine some features of the optimal tax. The main result is that the two kinds of social preferences mentioned above converge in order to grant tax exemption to incomes below the minimum wage, that is, a zero marginal tax rate for those incomes (the minimum wage is the income earned by those who work full time at the smallest wage rate, and an agent with any wage rate may earn less than this amount by working part time). But we also show under what circumstances it is legitimate to choose the tax that maximizes the minimum income (the transfer paid to agents who earn nothing), as advocated by promoters of the basic income proposal.

Our result of a zero marginal tax rate for the whole interval of incomes below the minimum wage is unusual in the optimal tax literature. In some cases, we cannot even exclude a negative marginal tax rate in this interval.

¹On double heterogeneity, see, for example, Seade (1979), dealing with commodity taxation, or Chone and Laroque (2005). The multidimensional screening problem has received much more attention in nonlinear pricing (see, e.g., Wilson 1993, Rochet and Chone 1998).

Some recent papers (Saez 2002, Choné and Laroque 2005) have obtained the possibility of zero or negative marginal tax rate by focusing on the case when agents either work full time or do not work at all. Our own result is not due to this assumption, since in our models agents have access to a continuum of labor quantities. Boadway et al. (2002) study a framework with double heterogeneity and continuous labor similar to ours and with a utilitarian social welfare function in which heterogeneous individual utility functions are given specific weights. They study a population with only two kinds of individual preferences and two skill levels, yielding four types of agents and three levels of incomes (they assume that “lazy” high skilled are indistinguishable from “crazy” low skilled). They obtain the possibility of a zero tax rate for the lowest income (and also of a negative tax rate for the highest income) only when the social objective gives such a high priority to hardworking agents that redistribution is made, regressively, from low incomes to high incomes. In contrast, our result is that low income earners not only have a zero marginal tax rate but also receive the greatest positive transfer among the whole population. We interpret our result as due to an assumption of a richer distribution of types (in particular, that there are agents with minimal productivity and sufficiently diverse preferences) and above all as due to social objectives that, as explained above, are neutral with respect to individual preferences and do not call for any redistribution among agents with identical productivity. We come back on the comparison between Boadway et al.’s (2002) result and ours at the end of Section 4.

As mentioned above, our ethical requirements are inspired by similar requirements studied in the literature on fairness (see Fleurbaey 1995). But that literature is devoted to determining the first-best allocations (for a recent survey, see Moulin and Thomson 1997) and ignores distortionary taxation. An important feature of fairness concepts is that they do not involve interpersonal *utility* comparisons but instead rely on comparisons of personal *resources* and how these resources are valued by individual noncomparable ordinal preferences.

On the other hand, the literature on optimal taxation usually relies, after Mirrlees (1971), on an additive social welfare function, which aggregates individual utilities (two recent, and quite synthetic, contributions have been made in Atkinson 1995 and Diamond 1998). The choice of individual utility functions is problematic for this approach, so that researchers often focus on results that are robust to the specification of the utility function. This difficulty is particularly acute when agents have heterogeneous preferences, as in Boadway et al. (2002) where weights are introduced in the social welfare function and results appear to be sensitive to how agents’ different utility functions are weighted. Our approach can be seen as supplementing the theory of optimal taxation by relating particular measures of individual well-being to fairness justifications and by showing that only a subset of the efficient taxation systems is compatible with the resulting social preferences.

As a matter of fact, there already is a small literature on equality of resources applied to optimal taxation. Indeed, our work elaborates on an early attempt made in Bossert, Fleurbaey, and Van de gaer (1999), which deals with linear tax. Related papers are by Roemer et al. (2000) and Schokkaert et al. (2004), who also focus on linear taxation. But none of these works rely on thoroughly constructed and axiomatically justified social preferences, and in particular, the social objectives used in Roemer et al. (2000) and Schokkaert et al. (2001) depend on the arbitrary choice of an interpersonally comparable representation of individual well-being, as in the welfarist literature (although they also consider “objective” indices that may not represent the agents’ personal preferences, in violation of the Pareto principle).

The paper is organized as follows. In Section 2, we present the model and notations. In Section 3, we introduce fairness principles and the related social preferences. In Section 4, we study the case where agents have either a low or a high skill and either a low or a high willingness to work, which illustrates the logic of the optimal tax in a simple way. In Section 5, we tackle the general case, and we derive the main results of the paper. In Section 6, we give some concluding comments. All the proofs are gathered in the Appendix.

2. The Model

There are two goods: labor and consumption. A *bundle* for agent i is a pair $z_i = (\ell_i, c_i)$, where ℓ_i is the labor and c_i is the consumption. The agents’ *consumption set* X is defined by the conditions $0 \leq \ell_i \leq 1$ and $c_i \geq 0$.²

The population contains n agents. Agents have two characteristics, their personal preferences over the consumption set and their personal skill, measured by their wage rate. For any agent $i = 1, \dots, n$, *personal preferences* are denoted by R_i and $z_i R_i z'_i$ (respectively, $z_i P_i z'_i$ and $z_i I_i z'_i$) means that bundle z_i is weakly preferred (respectively, strictly preferred, indifferent) to bundle z'_i . We assume that individual preferences are continuous, convex, and monotonic in the sense that for any $(\ell_i, c_i), (\ell'_i, c'_i) \in X$,

$$\ell_i \leq \ell'_i \text{ and } c_i > c'_i \Rightarrow (\ell_i, c_i) P_i (\ell'_i, c'_i).$$

We will say that preferences R_i exhibit a lower willingness to work than R'_i if they satisfy the following property: for any $(\ell_i, c_i), (\ell'_i, c'_i) \in X$,

$$\ell'_i > \ell_i \text{ and } (\ell'_i, c'_i) R_i (\ell_i, c_i) \Rightarrow (\ell'_i, c'_i) P'_i (\ell_i, c_i),$$

$$\ell'_i < \ell_i \text{ and } (\ell'_i, c'_i) R'_i (\ell_i, c_i) \Rightarrow (\ell'_i, c'_i) P_i (\ell_i, c_i).$$

This means that any pair of indifference curves, one for R_i and one for R'_i , cross at most once in the (ℓ, c) space, as illustrated in Figure 1, so that facing any kind of budget, agent R_i chooses a lower labor time than R'_i .

²Our analysis can easily be refined to take account of unequal time available for labor, when agents are constrained by family duties, for instance.

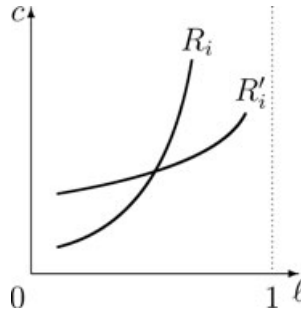


Figure 1: Indifference curves of different agents cross at most once in the (l, c) space.

For further reference it is useful to define the preferences R^c of agents who are only interested in their consumption (they have the highest willingness to work): For any $(\ell_i, c_i), (\ell'_i, c'_i) \in X$,

$$(\ell'_i, c'_i) R^c(\ell_i, c_i) \Leftrightarrow c'_i \geq c_i.$$

Agent i 's *skill* or *wage rate* is denoted by w_i and is measured in consumption units, so that w_i is the amount of consumption that agent i can afford when working $\ell_i = 1$ in the absence of tax or transfer. Wage rates are assumed to be fixed, as in a constant returns to scale technology. The maximal and minimal skills in the population are denoted as w_M and w_m , respectively.

An allocation $(z_1, \dots, z_n) \in X^n$ is feasible if $\sum_{i=1}^n c_i \leq \sum_{i=1}^n w_i \ell_i$. An agent i is said to be taxed (respectively, subsidized) at an allocation when $c_i < w_i \ell_i$ (respectively, $c_i > w_i \ell_i$).

In the first-best context, redistribution can be made via lump-sum transfers. Such transfers will be denoted as t_i (for agent i). This means that agent i 's first-best budget set $B^f(t_i, w_i)$ is then (see Figure 2 —on figures we adopt the convention that slopes of lines are indicated in parentheses below the lines):

$$B^f(t_i, w_i) = \{(\ell, c) \in X \mid c \leq t_i + w_i \ell\}.$$

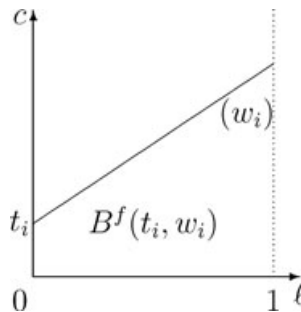


Figure 2: First-best budgets in the (l, c) space.

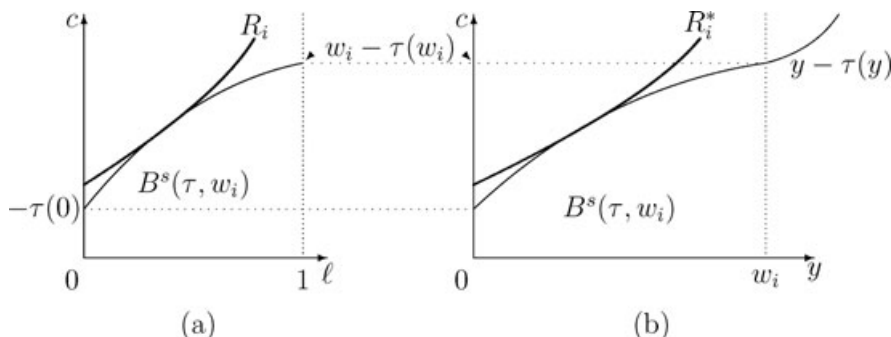


Figure 3: Second-best budgets (a) in the (l, c) space and (b) in the (y, c) space.

The budget constraint for the redistribution agency is then simply

$$\sum_{i=1}^n t_i \leq 0.$$

In the second-best context, we assume that only earned income $y_i = w_i \ell_i$ is observed, so that redistribution is made via a tax function $\tau(y)$. The tax may turn into a subsidy when $\tau(y) < 0$. Under this kind of redistribution, agent i 's budget is defined by (see Figure 3a)

$$B^s(\tau, w_i) = \{(\ell, c) \in X \mid c \leq w_i \ell - \tau(w_i \ell)\}.$$

It is convenient to focus on the earnings-consumption space, in which the budget is defined by (see Figure 3b; we retain the same notation as no confusion is possible)

$$B^s(\tau, w_i) = \{(y, c) \in [0, w_i] \times \mathbb{R}_+ \mid c \leq y - \tau(y)\},$$

and individual preferences R_i^* over income-consumption bundles can be defined, when $w_i > 0$, by³

$$(y, c) R_i^*(y', c') \Leftrightarrow \left(\frac{y}{w_i}, c\right) R_i\left(\frac{y'}{w_i}, c'\right). \quad (1)$$

The incentive compatibility constraint can then be formulated by the condition that for all agents i and j ,

$$(y_i, c_i) R_i^*(y_j, c_j) \text{ or } y_j > w_i.$$

The second part of the condition follows from our assumption that labor time is bounded above. We will illustrate its role in Section 4.

³When $w_i = 0$, then R_i^* is defined over $\{0\} \times \mathbb{R}_+$, and $(0, c) R_i^*(0, c')$ if and only if $c \geq c'$.

The budget constraint for the redistribution agency is

$$\sum_{i=1}^n \tau(w_i \ell_i) \geq 0.$$

We will say that a tax function τ is *feasible* if it satisfies the above constraint when all agents choose their labor time by maximizing their satisfaction over their budget set.

We restrict our attention to functions τ , which are continuous and such that $y - \tau(y)$ is nondecreasing.

3. Social Preferences

Help the low skilled or let the hardworking thrive? This alternative refers to two ethical principles, which we now proceed to formalize in the current model.

The reduction of inequalities due to unequal earning abilities should not be simply formulated in terms of disposable income, since efficiency alone would normally require a high-skilled agent to work more than a low skilled agent, and in terms of equity, working more may justify consuming more. Things are even more complex when preferences differ, too. But consider two agents with identical preferences and unequal skills. Then a reasonable objective is to avoid that they end up with bundles they both consider to be of unequal values, according to their identical preferences. In other words, the objective would be that they end up with bundles they deem equally valuable, that is, on the same indifference curve. Formulating such an objective in terms of a property that social preferences should satisfy, we require that reducing the indifference curve gap between two agents with identical preferences be a social improvement.

Help Low Skilled: It is a social improvement to change an allocation by modifying the bundles of two agents i and j who have identical preferences R , from z_i, z_j to z'_i, z'_j , so that

$$z_i P z'_i P z'_j P z_j.$$

The second ethical idea on which we focus here is that differences in preferences, in contrast to skills, may justify inequalities. In symmetry to the above principle, this is best applied to a pair of agents with identical skills, so as to avoid any issue of redistribution for unequal skills. If two agents have the same skill level, and the same budget, but one chooses to work hard while the other works less, this ethical view says that there is nothing objectionable about the resulting inequality. There are famous defenses of this kind of principle (e.g., Dworkin 1981, Rawls 1982), and they rely on the ideal of neutrality as regards individual preferences and choices.

This idea suggests that ideally redistribution should be made independently of preferences. That is, agents' preferences should never, by themselves, be a reason for redistributing resources. In a first-best context, it is actually possible to achieve this state of affairs by giving any agent i a lump-sum transfer t_i that does not depend on her preferences and letting her choose in the budget set $B^f(t_i, w_i)$. As a consequence, in the context of comparing allocations, it is interesting to focus on allocations that are obtainable via lump-sum transfers. Such allocations are those where every agent chooses her best bundle in a budget set of the $B^f(t_i, w_i)$ kind. The idea that redistribution should not depend on preferences clearly pushes in this case in the direction of reducing the inequality of budgets, and therefore the inequality in lump-sum transfers, for agents with equal skill. This is what the following condition states formally.⁴

Let Hardworking Thrive: It is a social improvement to change an allocation obtainable via lump-sum transfers by modifying the lump-sum transfers of two agents i and j who have the same skill, from t_i, t_j to t'_i, t'_j , so that

$$t_i > t'_i > t'_j > t_j.$$

Help Low Skilled and Let Hardworking Thrive can be directly related to the principles of "compensation" and "natural reward" coined in Fleurbaey (1995), where they are shown to conflict with each other. The two requirements formulated here are also mutually incompatible, as can be shown with a simple example. The Help Low Skilled principle focuses on indifference curves, whereas the Let Hardworking Thrive principle focuses on budgets. An agent with a low indifference curve compared to another may have a high budget with respect to a third one, and this may entail contradictory injunctions, as illustrated in Figure 4, where the Help Low Skilled principle says that agents 1 and 2 should be helped at the expense of agents 3 and 4, whereas the Let Hardworking Thrive principle advocates the contrary.

This conflict between principles implies that the social decision maker has to choose which of the two principles must have priority. Is it necessary to emphasize that this ethical dilemma is at the core of current debates about the welfare state? And it seems to go across the political spectrum, since one can read in the U.S. Democratic Party platform: "We must bring all Americans who are willing to work hard into the circle of prosperity." In the platform of the U.K. Labour Party, one finds, "Our values: (...) reward for hard work."⁵

⁴An interesting consequence of giving agents with equal skill the same budget is that they will not envy each other: $z_i R_i z_j$ for all i, j in a subpopulation with equal skill. This "no-envy among equally skilled" was the condition retained to express this ethical principle in Fleurbaey and Maniquet (1996) in a study of first-best allocation rules.

⁵See <http://www.democrats.org/issues/platform/platform.html#progress> and <http://www.labour.org.uk/>.

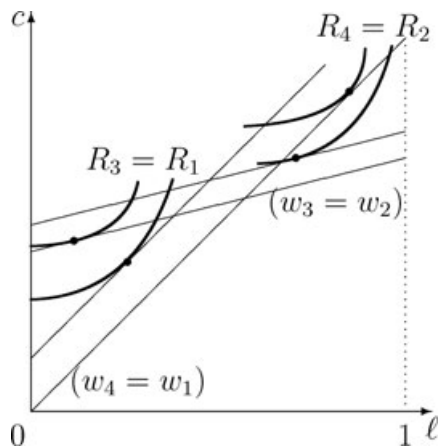


Figure 4: Social preferences cannot simultaneously satisfy Help The Low Skilled and Let Hardworking Thrive.

What we want to do here is to analyze the consequences of such ethical choices. Depending on which principle is preferred, a different kind of social preferences is obtained. Two kinds of social preferences are prominent in this context, as can be seen from the axiomatic analysis presented in Fleurbaey and Maniquet (2005). We here proceed directly with the presentation of these two kinds of social preferences.

We first have to define the notion of implicit budget associated with an agent's indifference curve and a wage rate w , which is the budget with a line of slope w that is tangent to the agent's indifference curve. Figure 5 illustrates this notion in the (ℓ, c) space (Figure 5a) and in the (y, c) space (Figure 5b).

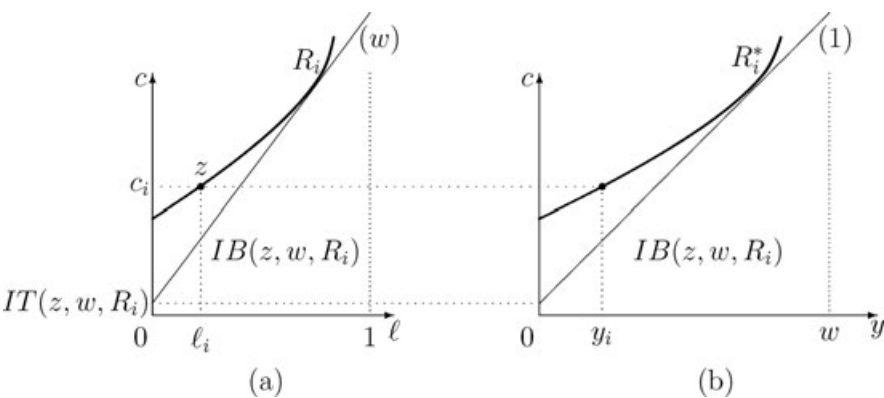


Figure 5: Implicit budgets (a) in the (ℓ, c) space and (b) in the (y, c) space.

The implicit budget of agent i at bundle z is formally defined as the set

$$IB(z, w, R_i) = B^f(t, w),$$

where t is computed as the minimal transfer such that there is $x \in B^f(t, w)$, xR_iz . Such a t will be called as the implicit transfer of agent i at bundle z , and denoted as $IT(z, w, R_i)$. This expression, viewed as a function of z , corresponds to a particular money-metric utility function. This measure of individual well-being does not require any information about individuals' subjective utility, and depends only on ordinal noncomparable preferences about consumption and labor.

The first social preferences focus on the implicit transfers with respect to a reference skill $\tilde{w}$. They are dubbed "egalitarian-equivalent" by reference to their similarity with Pazner and Schmeidler's concept (1978).

DEFINITION 1: (Egalitarian-equivalent social preferences EE). *Apply the maximin criterion to the vector*

$$(IT(z_1, \tilde{w}, R_1), \dots, IT(z_n, \tilde{w}, R_n)),$$

for some chosen $\tilde{w}$.

EE preferences give priority to the agents with the lowest $IT(z_i, \tilde{w}, R_i)$. As can be checked easily, EE preferences satisfy the Help Low Skilled principle and satisfy the Let Hardworking Thrive principle only for the special case of two agents having their identical skill precisely equal to $\tilde{w}$.

EE preferences are actually not uncommon in the optimal tax literature, and, for instance, Atkinson (1995) or Chone and Laroque (2005) study EE preferences with $\tilde{w} = 0$ (in this case, $IT(z_i, 0, R_i)$ is equal to the consumption level where agent i 's indifference curve crosses the axis). Usually the representation of individual well-being by $IT(z_i, 0, R_i)$ is presented as a more or less natural choice of utility function. In particular, when individual preferences are representable by a quasilinear utility function $c_i - v(\ell_i)$, such that $v(0) = 0$, one indeed has

$$c_i - v(\ell_i) = IT((\ell_i, c_i), 0, R_i).$$

What our approach provides is a rationale for the choice of $IT(z_i, \tilde{w}, R_i)$ as a measure of individual situations and an interpretation in terms of resources rather than subjective utility.

Since $IT(z_i, \tilde{w}, R_i)$ is a particular representation of i 's preferences, EE preferences satisfy the weak Pareto principle (an allocation where all agents are better off is socially strictly better).⁶ In the second-best context, one finds

⁶Fleurbaey and Maniquet (2005) actually characterize the leximin version of EE, that is, the lexicographic extension of the maximin (see, e.g., Sen 1986). The social preferences then satisfy the Strong Pareto principle.

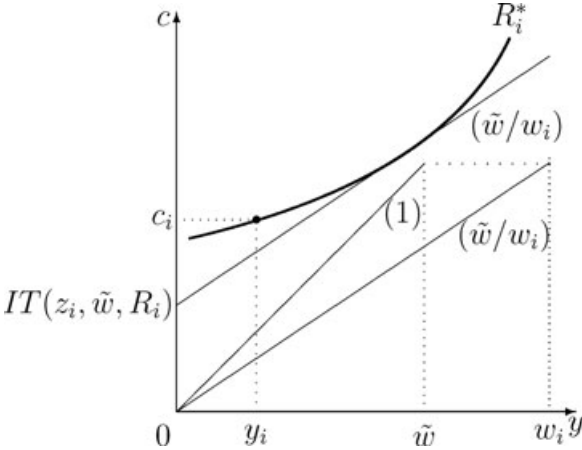


Figure 6: The construction of egalitarian-equivalent social preferences in the (y, c) space.

$IT(z_i, \tilde{w}, R_i)$ on the (y, c) space by taking the line of slope $\tilde{w}/w_i$, which is tangent to agent i 's indifference curve. The graphical construction is shown in Figure 6.

In the first-best context, the socially preferred allocation for EE preferences displays full equality of the $IT(z_i, \tilde{w}, R_i)$, and is described in Figure 7. It has all agents indifferent between their bundle and their best bundle in an equal budget with slope $\tilde{w}$. The bundles of resources that agents actually consume need not be on the boundary of this equal budget, as an agent's skill may differ from $\tilde{w}$.

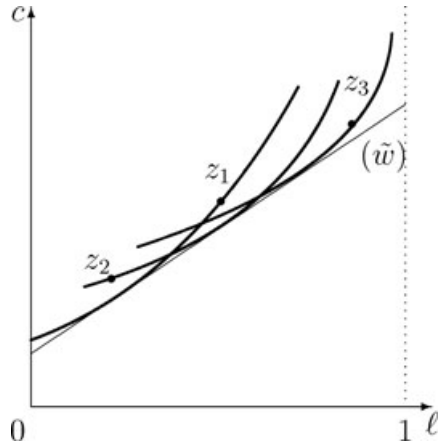


Figure 7: A first-best socially optimal allocation for egalitarian-equivalent social preferences.

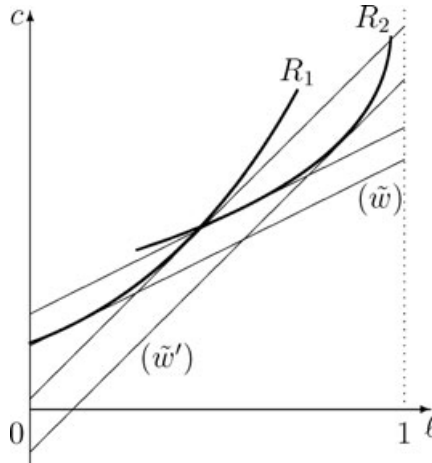


Figure 8: Two different reference parameter for egalitarian-equivalent social preferences.

EE preferences rely on the $\tilde{w}$ parameter. The consequences of the choice of this parameter are illustrated in Figure 8, where agent 1 has preferences that exhibit a lower willingness to work than agent 2. With a low $\tilde{w}$, EE preferences are relatively favorable to agent 1, because this agent is then considered to be the worse off, as measured by the implicit transfer $IT(z_i, \tilde{w}, R_i)$, and is therefore given priority by the maximin criterion. And symmetrically, a higher $\tilde{w}' > \tilde{w}$ entails that agent 2 is considered worse off in terms of $IT(z_i, \tilde{w}, R_i)$, and therefore relatively favored. In summary, EE preferences are neutral about preferences for agents with $w_i = \tilde{w}$, are biased toward the “hardworking” for agents with $w_i < \tilde{w}$, and conversely for $w_i > \tilde{w}$.

For the definition of the second kind of social preferences, we introduce the following convention. If U is a continuous utility function defined over X and $B \subset X$ is a compact subset of bundles (such as a budget set), then

$$U^{\max}(B) = \max\{U(z) \mid z \in B\}.$$

DEFINITION 2: (Equivalent-budget social preferences EB). *Apply the maximin criterion to the vector*

$$(\tilde{U}^{\max}(IB(z_1, w_1, R_1)), \dots, \tilde{U}^{\max}(IB(z_n, w_n, R_n))),$$

where $\tilde{U}$ is any utility function representing some chosen preferences $\tilde{R}$.

EB preferences give priority to the agents with the worst $IB(z_i, w_i, R_i)$ according to $\tilde{R}$. The notion of implicit budget was illustrated on Figure 5.

EB preferences satisfy the Let Hardworking Thrive principle, but satisfy the Help Low Skilled principle only for the special case of two agents having their identical preferences equal to $\tilde{R}$.

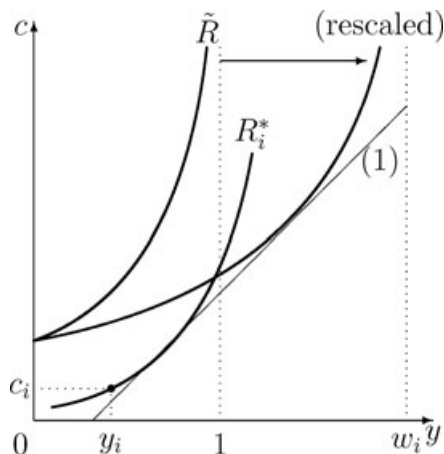


Figure 9: The construction of equivalent-budget social preferences in the (y, c) space.

Notice that the expression $\tilde{U}^{\max}(IB(z_i, w_i, R_i))$ is increasing with i 's satisfaction. Therefore EB preferences satisfy the Pareto principle, and this guarantees that fairness requirements will never be allowed to entail inefficiency in the chosen allocations.

In the second-best context, one finds $IB(z_i, w_i, R_i)$ in the (y, c) space by looking for the line of slope 1, which is tangent to agent i 's indifference curve. In order to evaluate this implicit budget with $\tilde{R}$, one has to rescale $\tilde{R}$ to agent i 's consumption set in this space, as explained in Equation (1) and shown in Figure 9.

In the first-best context, EB preferences lead to a fully egalitarian situation, in the sense that all the agents' implicit budgets are deemed equivalent by preferences $\tilde{R}$, as shown in Figure 10.

EB preferences rely on the $\tilde{R}$ preferences as a crucial parameter. The choice of this parameter has nothing to do with favoring agents who have such preferences, but only with guaranteeing that the Help Low Skilled principle will be fully satisfied for them. Practically, the choice of $\tilde{R}$ influences the degree of redistribution. Examine Figure 11, where agent 1 has a smaller skill $w_1 < w_2$, and two implicit budgets are represented (we do not represent preferences of agents 1 and 2 as they would not influence the degree of redistribution). Preferences $\tilde{R}$ exhibit a lower willingness to work than $\tilde{R}'$. Choosing $\tilde{R}$ will lead to little redistribution in favor of agent 1, as agent 2's implicit budget is considered the worse of the two. In contrast, choosing $\tilde{R}'$ will entail a high degree of redistribution in favor of agent 1, because implicit budgets are now assessed by comparing consumption levels associated with larger labor time, so that the implicit budget of the low wage rate agent (agent 1) is deemed worse than that of the high-skilled agent.

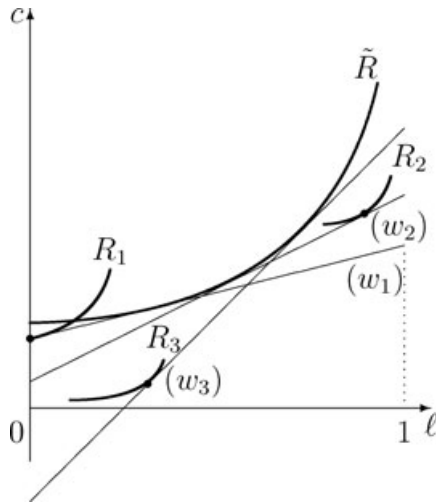


Figure 10: A first-best socially optimal allocation for equivalent-budget social preferences.

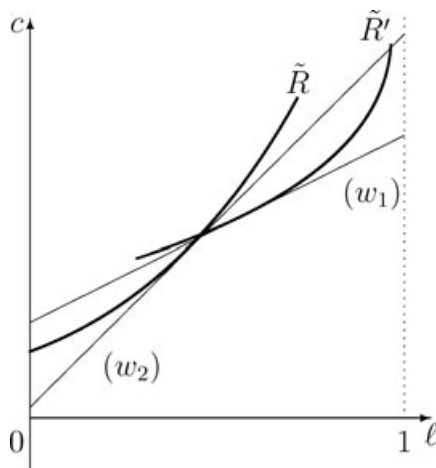


Figure 11: Two different reference preferences for equivalent-budget social preferences.

4. Optimal Tax: The Two-by-Two Case

In the previous section, we have described social preferences and their consequences in the first-best context. We now turn our attention to the second-best context, and start with a preliminary analysis of the case where skills can be either low or high and preferences exhibit either a low or a high willingness to work. More precisely, there are two possible skills, $w_1 < w_2$, and two possible

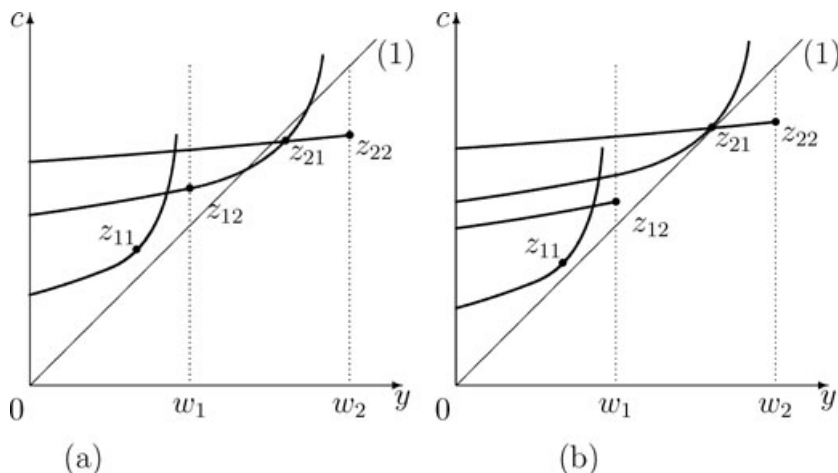


Figure 12: Incentive compatible and (second-best) efficient allocations (a) where $z_{21} l_{21}^* z_{12}$ and (b) where $z_{21} p_{21}^* z_{12}$.

preferences (in the (ℓ, c) space), R_1 and R_2 such that R_1 exhibits a lower willingness to work than R_2 (i.e., the Mirrlees–Spence single crossing condition holds in the (ℓ, c) space among agents with the same skill).

Furthermore, we assume that low-skill high willingness to work agents cannot be distinguished from high-skill low willingness to work ones. That is, let $R_i^*|_A$ denote the restriction of preferences R_i^* to a subset A of bundles. Then, $R_{12}^* = R_{21}^*|_{[0, w_1] \times \mathbb{R}_+}$, where the first index refers to skill and the second one to preferences. To fix ideas (in particular for graphical illustrations), we assume that there is one agent of each type in the population.

Let us begin by illustrating the influence of our assumption that labor time is bounded above on the incentive constraints. Figure 12 depicts two incentive compatible and (second-best) efficient allocations where it is not true that $z_{12} R_{12}^* z_{21}$, but $y_{21} > w_1$. In part (a) of the picture, $z_{21} l_{21}^* z_{12}$, whereas in part (b) $z_{21} p_{21}^* z_{12}$ (by incentive compatibility, we cannot have $z_{12} p_{21}^* z_{21}$). Let us observe that both parts are consistent with our assumption that $R_{12}^* = R_{21}^*|_{[0, w_1] \times \mathbb{R}_+}$. In this case, the constraint $l_{12} \leq 1$ prevents the planner from proposing bundle (y_{21}, c_{21}) to agents 12 and benefiting from the fact that $\tau(y_{21}) > \tau(y_{12})$.

To simplify the exposition and to be able to compare our results with the literature, we now add the assumption that $z_{12}^* = z_{21}^*$.⁷ Consequently, we can do as if there were three groups of agents. A group of agents with low skill and low willingness to work, a group with high skill and high willingness to work, and a mixed group gathering the other agents (let us note that the

⁷This could follow, for instance, from the assumption that agents 12 never choose $l_{12} = 1$ when their marginal after-tax income is lower or equal to w_1 .

single crossing condition then also holds in the (y, c) space). An allocation $(y_{ij}, c_{ij})_{i,j \in \{1,2\}}$ is incentive compatible if $(y_{12}, c_{12}) = (y_{21}, c_{21})$ and

$$(y_{11}, c_{11}) R_{11}^*(y_{21}, c_{21}), (y_{12}, c_{12}) R_{12}^*(y_{22}, c_{22}), \\ (y_{22}, c_{22}) R_{22}^*(y_{12}, c_{12}), (y_{21}, c_{21}) R_{21}^*(y_{11}, c_{11}).$$

The first two conditions are upward conditions (no agent should be willing to mimic an agent with the same willingness to work but a higher skill), and the last two are downward conditions (no agent should be willing to mimic an agent with the same willingness to work but a lower skill).

Four social preferences are of particular interest in this model, the *EE* preferences with either w_1 or w_2 as reference skill, and the *EB* preferences with either R_1 or R_2 as reference preferences. We can derive the following preliminary results. The first result states that redistribution between high willingness to work agents must go from high-skilled to low skilled agents, with three out of our four social preferences.

PROPOSITION 1: *The incentive constraint $(y_{22}, c_{22}) R_{22}^*(y_{12}, c_{12})$ is binding in all cases but when the social preferences are $R_1 - EB$.*

The second proposition, illustrating our main result in the two-by-two case, proves that the social preferences derived from our two conflicting principles of Help Low Skilled and Let Hardworking Thrive may lead to similar recommendations in the second best.⁸

PROPOSITION 2: *An incentive compatible allocation $(y_{ij}, c_{ij})_{i,j \in \{1,2\}}$ is optimal for social preferences $w_1 - EE$ if and only if it is optimal for social preferences $R_2 - EB$.*

Optimal allocations are illustrated in Figure 13. In Figure 13(a), $(z_{11}, z_{12} = z_{21}, z_{22})$ is optimal for social preferences $w_1 - EE$ and $R_2 - EB$. The argument can be developed geometrically. First, the indifference curves of both types of low-skilled agents are tangent to the same budget of slope 1. This implies that $IB((\ell_{11}, c_{11}), w_1, R_1) = IB((\ell_{12}, c_{12}), w_1, R_2)$, so that the $w_1 - EE$ and $R_2 - EB$ well-being indices are equalized among low-skilled agents. Moreover, the budget of slope $\frac{w_1}{w_2}$ represents the same implicit budget rescaled so as to fit the consumption set of high-skilled agents. Given that indifference curves of agents 22 can be obtained by a corresponding rescaling of those of agents 12, the incentive compatibility constraint $z_{22} R_{22}^* z_{12}$ guarantees that $IT((\ell_{22}, c_{22}), w_1, R_2) \geq IT((\ell_{12}, c_{12}), w_1, R_2)$, so that the indifference curve of agents 22 must be above the line of slope $\frac{w_1}{w_2}$ (illustrating Proposition 1, as equalizing the well-being measures is impossible among high willingness to work agents). Consequently, low-skilled agents have the lowest $w_1 - EE$

⁸In the case of only two agents, optimal allocations for $w_2 - EE$ and $R_1 - EB$ also coincide.

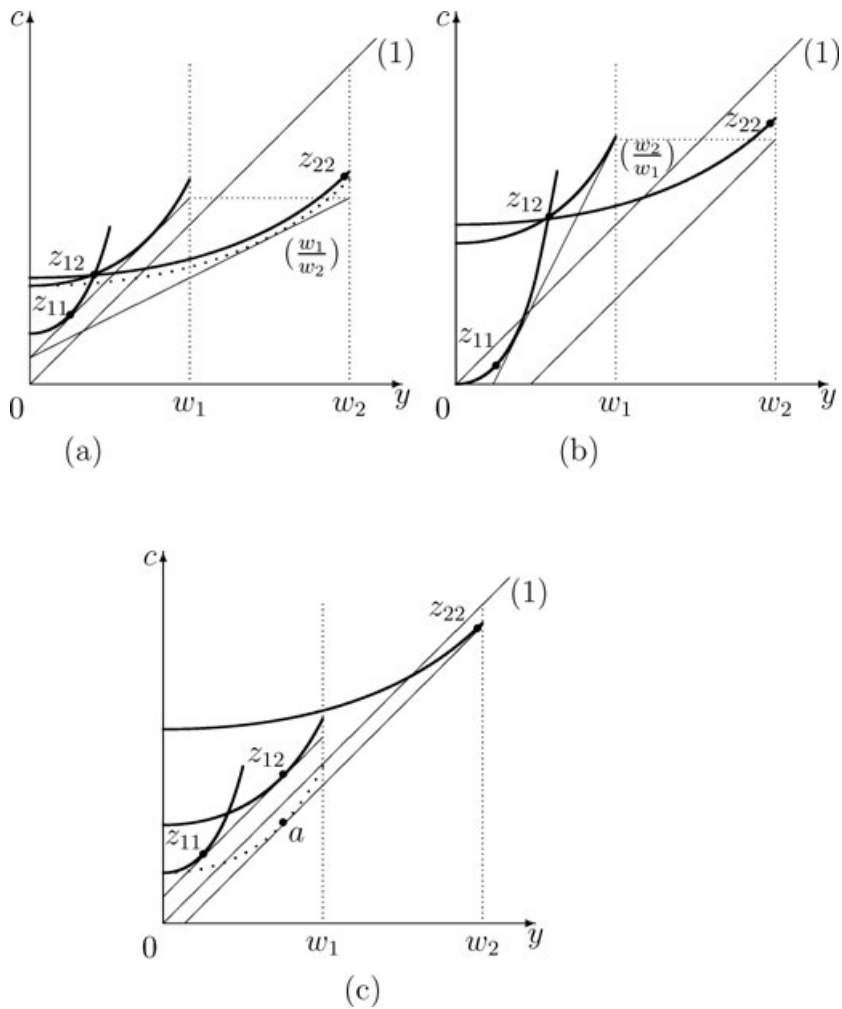


Figure 13: Optimal second-best allocations (a) for social preferences w_1 -EE and R_2 -EB, (b) for social preferences w_2 -EE, and (c) for social preferences R_1 -EB.

well-being indices. Also, the dashed indifference curve is a rescaling of the appropriate indifference curve of the reference agent (an agent with preferences R_2) and, as explained above, this curve lies below the indifference curve of agents 22 through z_{22} , so that, again, low-skilled agents have the lowest $R_2 - EB$ well-being indices. Now, the incentive compatibility constraints are binding in such a way that it is no longer possible to increase the welfare of any of the low-skilled agents, proving that the allocation is $w_1 - EE$ and $R_2 - EB$ optimal.

In Figure 13(b), $(z_{11}, z_{12} = z_{21}, z_{22})$ is optimal for social preferences $w_2 - EE$. Again, the indices are equalized among low-skilled agents, which is

illustrated by the fact that both indifference curves are tangent to the same line of slope $\frac{w_2}{w_1}$, and the welfare of no low-skilled agent can be increased without violating an incentive compatibility constraint. Let us observe that, as suggested by the figure, it could be the case that $(0, 0) P_{11}^* z_{11}$, that is, agents 11 would be forced to work.

In Figure 13(c), $(z_{11}, z_{12} = z_{21}, z_{22})$ is optimal for social preferences $R_1 - EB$. First, the reference agent (an agent with preferences R_1) is indifferent between the implicit budgets of all low-skilled agents (as they all have their indifference curves tangent to the same 45° line). Secondly, the rescaled indifference curve matching preferences R_{21}^* is tangent (at point a) to agents' 22 implicit budget, which means that the relevant index has been equalized among agents of all three relevant groups (agents 21 have a higher index level but this cannot be avoided as they cannot be distinguished from agents 12). Let us observe that, in this example, the allocation is also first-best efficient.

Those figures illustrate the effect of the choice of w_i or R_i on the resulting optimal allocation. It confirms what we said about first-best optimal allocations. When social preferences are of the *EE* type, a larger reference skill benefits the low-skilled hardworking agents (agents 12). When social preferences are of the *EB* type, a lower willingness to work of the reference preference benefits the high-skilled hardworking agents (and hurts all the other agents). We see that *EE* preferences lead to more redistribution than *EB* preferences, but the difference is not as large as could be expected given that, as proven by Proposition 2, the sets of optimal allocations intersect.

The taxation is never progressive among low-skilled agents. Indeed, the amount transferred to low-skilled hardworking agents is always at least as large as the amount transferred to low-skill low willingness to work agents (those transfers are measured by the vertical distance between the bundles agents are assigned and the no tax 45° line). Moreover, in two cases, with $w_1 - EE$ and $R_2 - EB$ social preferences (see Figure 13a), it is strictly larger. We need to qualify this result, however. With $w_1 - EE$ preferences, the difference in the amount received by agents 12 and 11 depends on the shape of the indifference curve of agents 11, given that the incentive compatibility constraint $z_{11} R_{11}^* z_{21}$ is binding. Had they been linear with a slope equal to 1, both groups would have received the same transfer (bundle z_{12} would have been on the frontier of the same implicit budget set as z_{11}). As we will see in the next section, when the number of agents is large, preferences are sufficiently diverse, and the reference skill is the lowest skill in the population, low-skilled agents indeed receive the same transfer. With another reference skill, in our case with w_2 , the transfer paid to agents 12 is strictly larger than the one paid to agents 11 independently of the shape of the preferences. The resulting tax system is therefore regressive for low incomes (the marginal rate of tax is, on average, negative). This result will still hold with a large number of agents.

All the tax systems we justify with *EE* or *EB* social preferences are progressive if we regard larger incomes: The tax paid by agents 22 is always larger than the one paid by any other agent. This result, however, does not extend to a larger population.

Finally, let us observe that, even if our social preferences are of the maximin type, they may lead to a very limited redistribution, as exemplified by the $R_1 - EB$ preferences. The intuition for this result is the following one. The well-being index associated with *EB* preferences is equalized among agents when an agent with reference preferences is indifferent between the implicit budgets of all agents, and, in particular, is indifferent between having a low or a high skill. But the transfer that is necessary for an agent with a low willingness to work to be indifferent between budgets corresponding to a low and a high skill is lower than the transfer necessary for an agent with a high willingness to work. At the limit, an agent with so low a willingness to work that she always chooses not to work (she has indifference curves with slope greater than w_M) does not need any transfer to be indifferent between having a low or a high skill. Consequently, if these preferences are the reference ones, then the laissez-faire is optimal.

Boadway et al. (2002) study the same two-by-two model under the assumption that preferences are quasilinear in leisure and there is no upper bound on the labor time. They show that when social preferences are utilitarian, the optimal allocations can be associated with a list of binding incentive compatibility constraints very different from what we obtain here. In particular, it may be the case that redistribution benefits the 22 agents. This cannot occur here as those agents always have larger index of well-being and we apply the maximin criterion. This proves that allocations maximizing one of the social preferences discussed in this paper form a strict subset of the second-best efficient allocations.

Closer to what we develop here, they study the consequences of a family of utilitarian social preferences which give absolute priority to low-skilled agents in the sense that the weight assigned to the others' utility is zero, whereas the weights given to the utilities of low and high willingness to work agents, among the low-skilled, may vary. Let us call those functions utility maximin functions. Boadway et al. (2002) also reach the conclusion that redistribution may be regressive among low-skilled agents. Actually, the set of allocations which are optimal for one of their utility maximin functions coincide with the set of optimal allocations we would obtain with $\tilde{w} - EE$ social preferences by letting $\tilde{w}$ vary over $\mathbb{R}_+$. Indeed, by a slight generalization of our study of $w_1 - EE$ and $w_2 - EE$ above, we can show that if an allocation is incentive compatible, the low-skilled agents are always those having the lowest well-being index, so that maximizing one of those social preferences boils down to maximizing a utilitarian social welfare function where the weights assigned to high-skilled agents are zero. Moreover, if we take $\tilde{w} = 0$, the $\tilde{w} - EE$ force us to maximize the welfare of agents 11, and, as $\tilde{w}$ increases, the $\tilde{w} - EE$ objective becomes that of maximizing a utilitarian social welfare function where the weight of

11 agents decrease and that of 12 agents increase (and the other weights remain zero). This equivalence between our $\tilde{w} - EE$ preferences and the Boadway et al. maximin only holds, however, when there are two kinds of individual preferences in the population. As shown in the next section, in the general case, $\tilde{w} - EE$ preferences will still lead to allocations having the property that the indifference curves of all low-skilled agents are tangent to the same line of slope $\frac{\tilde{w}}{w_m}$, whereas utility maximin social preferences may select other allocations.

Let us complete this section by mentioning a last feature of the optimal allocations identified here. Recall that agents 12 and 21 are indistinguishable in the (y, c) space. Consequently, as soon as an allocation $(z_{11}, z_{12} = z_{21}, z_{22})$ is second-best efficient, it remains so independently of how the total measure of agents 12 and 21 is distributed among the two types. Moreover, given that our objectives are egalitarian, and our indices of well-being are indeed equalized among agents 11 and 12, optimality of the selected allocations is preserved after a resizing of subgroups 12 and 21 that does not affect their total size (with an obvious discontinuity when the proportion of agents 12 reaches 0). This contrasts with the outcome of a utilitarian objective where an increase in the relative size of a group of agents always benefits these agents.

5. Optimal Tax: The General Case

We now consider the case of a general population. What we do in this section is an analysis of how the search for the optimal tax function can be translated into simple objectives about the shape of the function of disposable income, and we derive some conclusions about the shape of the optimal tax function.

5.1. The Case $w_m = 0$

Let us focus, first, on the case when $w_m = 0$, and let us assume moreover that there are agents with a zero skill, for every type of preferences R in the population.⁹ For any tax function, the budget set of such agents in the (ℓ, c) space is the flat line with intercept $-\tau(0)$. In every preference class, these agents have the lowest budget, hence the lowest indifference curve, and therefore the lowest $IT(z_i, \tilde{w}, R_i)$.

Therefore the optimal tax for EE preferences must maximize the welfare of agents with zero skill. Such agents have heterogeneous preferences, which may a priori render this maximization ambiguous, but fortunately here, maximizing their welfare is, for all kinds of individual preferences R , tantamount to maximizing $-\tau(0)$.

⁹For instance, the distributions of w and R are independent, and the minimum value for w is zero.

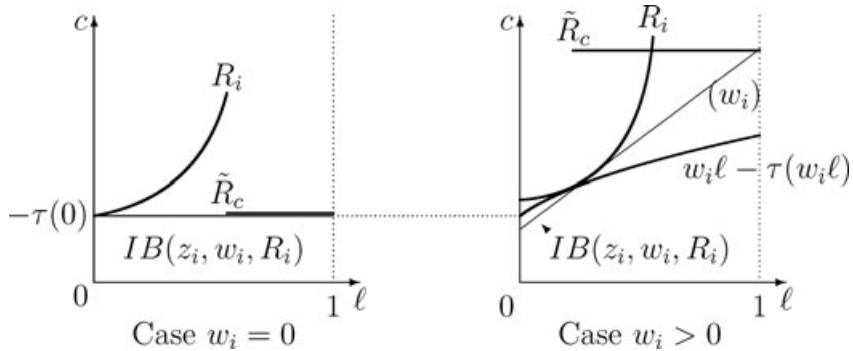


Figure 14: The equivalent-budget social preferences with reference preferences $\tilde{R} = R^c$.

A similar reasoning can be made for *EB* preferences based on the highest willingness to work preferences $\tilde{R} = R^c$. For any tax function, the agents with zero skill necessarily have the following implicit budget:

$$\{(\ell, c) \mid 0 \leq c \leq -\tau(0)\}.$$

And this is always the worst implicit budget for preferences R^c , because all other agents are at least as well off as at the bundle $(0, -\tau(0))$, and therefore their implicit budget must have a maximal consumption that is greater than or equal to $-\tau(0)$, as shown in Figure 14.¹⁰

We have therefore proved the following result.

THEOREM 1: *If every preference class contains agents with zero skill, then the optimal tax for all EE preferences, and for EB preferences with $\tilde{R} = R^c$, maximizes the minimum income $-\tau(0)$.*

When the reference preferences are different from R^c , then *EB* preferences lead to tax systems which are similar to those obtained when $w_m > 0$. We come back to those cases in Section 5.3.

5.2. EE Tax When $w_m > 0$

Let us now examine the case when the minimum skill in the population is positive, and start by focusing on *EE* preferences, for the case $\tilde{w} \geq w_m$.

The optimal tax is complex to analyze when there is no restriction on the distribution of skills and preferences. Here, we assume that pretax income is not too informative a signal in the following sense: It never happens that over some interval of income $[y, y']$ with $0 < y < y' \leq w$, one finds only agents with wage greater than w , whereas agents with wage equal to w can

¹⁰Notice that here we only need the assumption that there is at least one agent in the population with a zero wage.

be found earning incomes either below y or above y' . We then impose the assumption that over any interval of income $[0, w]$, it is impossible, by only looking at preferences restricted to that interval of income, to identify agents with greater productivity than w and distinguish them, on the basis of their preferences, from agents with productivity w . Let W denotes the set of wage rates in the current profile:

$$W = \{w \in \mathbb{R}_+ \mid \exists i, w_i = w\}.$$

ASSUMPTION (No Identification): *For every agent i , and every $w \in W$ such that $w < w_i$, there is another agent j such that $w_j = w$ and $R_j^* = R_i^*|_{[0, w] \times \mathbb{R}_+}$.*

This assumption is a generalization of the assumption made in the previous section, where agents 12 and 21 could not be distinguished. Let us insist that even if No Identification is a strong assumption it is only imposed to exclude the implausible case where fiscal authorities design the tax for some interval $[y, y']$ (and, in particular, apply very high tax rates over that interval) only because they know that nobody among agents having an ability to earn income in the interval $[0, y'']$ for some $y'' > y'$ would ever end up with an income in $[y, y']$.

We obtain the following description of the optimal tax, when it exists. Let $a^+ = \max\{a, 0\}$.

THEOREM 2: *Under the No Identification assumption, a tax function maximizing*

$$\min_i IT(z_i, \tilde{w}, R_i),$$

for $\tilde{w} \geq w_m$, can be computed by minimizing $\tau(w_m)$ under the constraint: for all $y \geq 0$,

$$\tau(w_m) \leq \tau(y) \leq \tau(w_m) + y - w_m + (w_m - y)^+ \left(\frac{\tilde{w}}{w_m} \right).$$

Theorem 2 is illustrated in Figure 15: In the income–consumption space, computing the optimal tax amounts to maximizing the height of the point $(w_m, w_m - \tau(w_m))$ under the constraint that the income function $y - \tau(y)$ must be located in the area delineated by the thick lines.

An income function in this area has the following features. First, the agents who earn w_m receive the greatest subsidy, in absolute amount. Secondly, the average marginal tax rate over incomes belonging to the interval $[0, w_m]$ is nonpositive. And in the particular case when $\tilde{w} = w_m$, which is, as explained above, the most generous to low incomes, the lower bound coincides with the upper bound for this interval, so that the marginal tax rate is exactly zero over $[0, w_m]$.

The intuitive argument for this zero marginal tax result goes as follows. Consider any incentive compatible allocation $z = (z_1, \dots, z_n)$, and let the

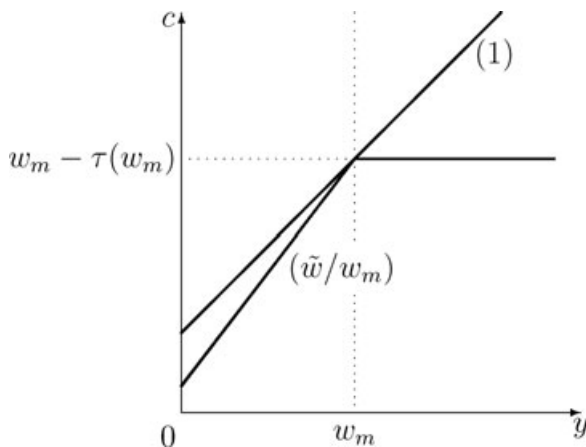


Figure 15: Lower and upper bounds on the optimal tax scheme for egalitarian-equivalent social preferences.

corresponding tax function τ be defined such that $y - \tau(y)$ is the envelope curve of the agents' indifference curves (in (y, c) space) at z . Consider the family of functions f_α defined by $f_\alpha(y) = \alpha + \min\{y, w_m\}$. There is a maximal α such that for all y , $y - \tau(y) \geq f_\alpha(y)$. Since $y - \tau(y)$ is nondecreasing, necessarily there is $y \in [0, w_m]$ such that $y - \tau(y) = f_\alpha(y)$.

In words, the graph of f_α touches the envelope curve of the agents' indifference curves on $[0, w_m]$, as illustrated in Figure 16. But, by the No Identification assumption, this part of the envelope curve is also the envelope curve of the subpopulation of agents with productivity w_m . Therefore, at z ,

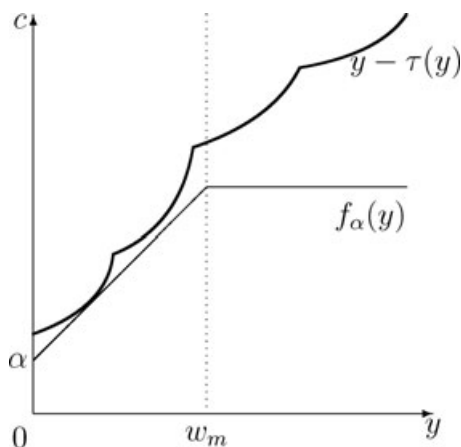


Figure 16: The graph of f_α , touches the envelope curve of the agents' indifference curves on $[0, w_m]$.

there is an agent i , with productivity w_m , such that (remember, indeed, how the implicit budget is computed in (y, c) space)

$$IT(z_i, \tilde{w}, R_i) = \alpha.$$

It is easy to check, by the same token, that for all other agents j of the whole population,

$$IT(z_j, \tilde{w}, R_j) \geq \alpha.$$

This is the first part of the argument, allowing us to identify the bundle of the worst-off agent in the economy. The second part is that if one truncates the tax function τ and replaces it with $\tau_0(y) = \max\{\tau(y), -\alpha\}$, one obtains a new allocation that is still feasible. Indeed, τ_0 is obtained from τ by cutting all subsidies greater than α .¹¹ The change does not affect agents who received less under z (or were taxed), because it cuts a part of their budget which they did not choose anyway, and for all affected agents, the subsidy they receive afterward is smaller. It is then an immediate accounting result that the new tax function is feasible. But it still satisfies $y - \tau_0(y) \geq f_\alpha(y)$ for all y , and therefore, for all i in N , one still has

$$IT(z_i, \tilde{w}, R_i) \geq \alpha.$$

The important point is that, for all $y \leq w_m$, $y - \tau(y) \geq f_\alpha(y)$ means that $\tau(y) \leq -\alpha$, so that $\tau_0(y) = -\alpha$, implying that τ_0 has a zero marginal tax rate over $[0, w_m]$. And this is why the optimal tax may be sought among the subfamily of tax functions displaying this property (see the formal proof of the theorem, for more details). This result does not prove that all taxes maximizing $\min_i IT(z_i, \tilde{w}, R_i)$ have this shape. But the theorem proves that there is no loss of social welfare if one restricts his attention to those taxes.

5.3. EB Tax When $w_m \geq 0$

Let us now turn to EB preferences, for the general case $w_m \geq 0$. It is not obvious to find the minimal $\tilde{U}^{\max}(IB(z_i, w_i, R_i))$ over the whole population, because a higher skill does not necessarily mean a larger implicit budget $IB(z_i, w_i, R_i)$, since budget frontiers may intersect. In a given subpopulation with skill w , however, a part of the reasoning leading to Theorem 2 still applies, that is, the smallest implicit budget is obtained by the line of appropriate slope that is tangent (from below) to the envelope curve of those agents' indifference curves. In the (ℓ, c) space (Figure 17a), this is the line of slope w . In the (y, c) space (Figure 17b), the slope is 1, and the budget is constrained by $y \leq w$.

¹¹We can restrict our attention to allocations such that the maximal α is nonnegative, since the laissez-faire allocation yields $\alpha \geq 0$. Therefore, the allocation that maximizes $\min_i IT(z_i, \tilde{w}, R_i)$ has $\alpha \geq 0$.

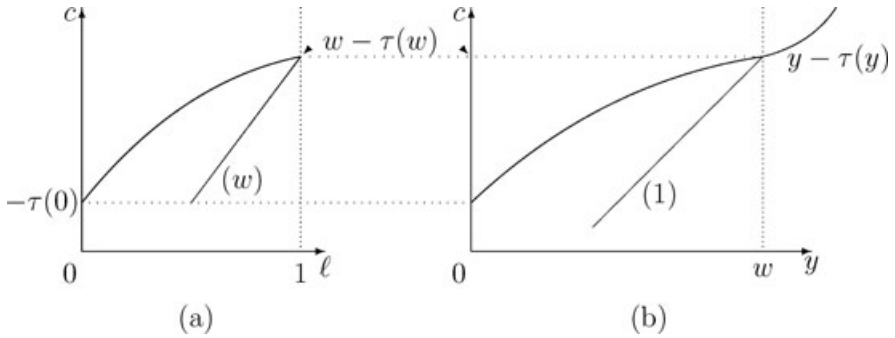


Figure 17: Identifying the smallest implicit budget (a) in the (l, c) space and (b) in the (y, c) space.

It is worthwhile noting here that if a tax function τ is nondecreasing (a case that will be shown to be relevant below), then $y - \tau(y)$ has a slope less than or equal to 1, so that the smallest implicit budget among the subpopulation with skill w is obtained by agents who work full time (as shown in Figure 17b), if there are such agents.

Define the expenditure function $\tilde{E}(w, u)$, relative to reference preferences $\tilde{R}$, as the solution to the equation

$$\tilde{U}^{\max}(B^f(\tilde{E}(w, u), w)) = u.$$

THEOREM 3: Under the No Identification assumption, a tax function maximizing

$$\min_i \tilde{U}^{\max}(IB(z_i, w_i, R_i))$$

can be computed by minimizing $\tau(w_m)$ under the constraint: for all $y \geq 0$,

$$\tau(w_m) \leq \tau(y) \leq \min_{w \in W} [(y - w)^+ - \tilde{E}(w, \tilde{U}_0)],$$

where $\tilde{U}_0 = \tilde{U}^{\max}(B^s(-\tau(w_m), w_m))$.

The inequalities obtained in the theorem yield upper and lower bounds for the income function $y - \tau(y)$, which are represented by the thick lines on Figure 18. Again, one observes that the agents who earn $y = w_m$ receive the greatest subsidy over the whole population. And they share this privilege with all agents with a lower income, since the marginal tax rate over the interval $[0, w_m]$ is exactly zero.

On the figure, w_m , w , w' , and w'' are consecutive members of W . The intuition for the construction of the upper bound on the tax goes as follows. Within each group of agents having the same wage rate, if the tax is nondecreasing, those working full time have the lowest well-being index (see Figure 17). Given that the social preferences are of the maximin type, we can restrict our attention to those agents. The upper bound is indeed designed

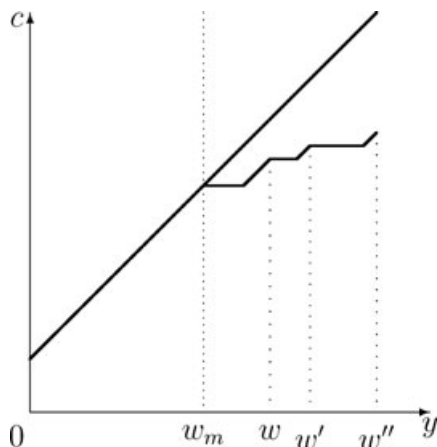


Figure 18: Lower and upper bounds on the optimal tax scheme for equivalent-budget social preferences.

in such a way that the well-being index is equalized among agents working full time.

This is illustrated in Figure 19: For any $w \in W$, the implicit budget of an agent choosing a maximal labor time is equal to $B^f(-\tau(w), w)$, which are referred to B_m , B' , and B'' in the figure, for wages w_m , w' , and w'' , respectively. Equality of $\tilde{U}^{\max}(B^f(-\tau(w), w))$ is achieved among w 's when the rescaled indifference curves $\tilde{I}_m$, $\tilde{I}'$, and $\tilde{I}''$ associated with the same welfare level (they

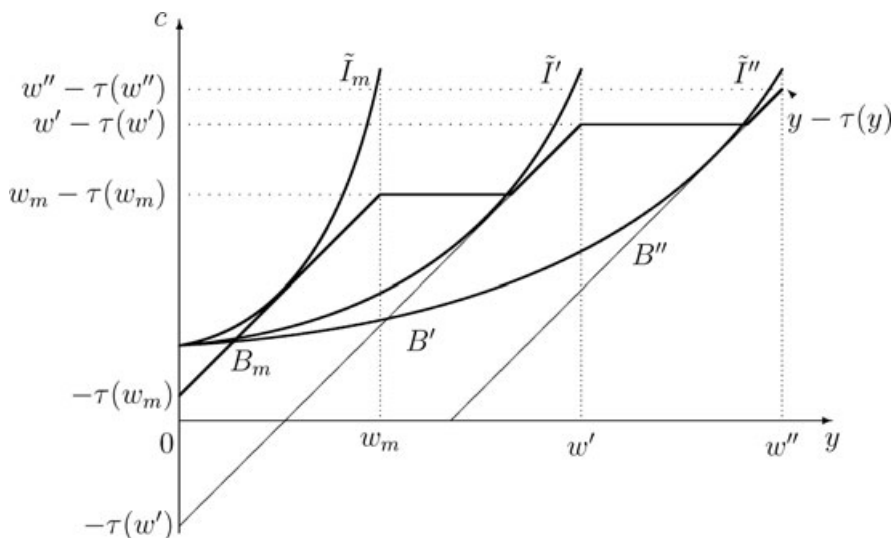


Figure 19: Identifying the upper bound on the tax scheme for equivalent-budget social preferences.

intersect the vertical axis at the same coordinate) are tangent to B_m , B' , and B'' , respectively.

If we fix $\tilde{U}_0 = \tilde{U}^{\max}(B^f(-\tau(w_m), w_m))$, we need to have $-\tau(w) = \tilde{E}(w, \tilde{U}_0)$. Then, for any y between, say, w' and w'' , the tax $\tau(y)$ must satisfy $y - \tau(y) \geq w' - \tau(w')$ and $\tau(y) \leq \tau(w'')$, and the definition of the upper bound follows.

Moreover, for $y \in [0, w_m]$, the only relevant wage is w_m and the restriction on the tax is $\tau(y) \leq \tau(w_m)$. Combined with the lower bound $\tau(y) \geq \tau(w_m)$, we obtain $\tau(y) = \tau(w_m)$ which means that the marginal income tax rate, again, needs to be zero on low incomes.

To study the shape of the function

$$\min_{w \in W} [(y - w)^+ - \tilde{E}(w, \tilde{U}_0)],$$

we can focus, for the sake of simplicity, on the case when W is the interval $[w_m, +\infty[$. For a given y , the expression $(y - w)^+ - \tilde{E}(w, \tilde{U}_0)$ is nondecreasing in w , so that the minimum is attained for $w = y$ if $y \in W$, and for $w = w_m > y$ otherwise. Then one can write

$$\min_{w \in W} [(y - w)^+ - \tilde{E}(w, \tilde{U}_0)] = -\tilde{E}(\max\{y, w_m\}, \tilde{U}_0).$$

This expression is constant for $y \leq w_m$, and is convex because $\tilde{E}(w, \tilde{U}_0)$ is concave in w . In other words, this function provides an upper bound for the tax function, which has nondecreasing marginal tax rates, a clear case of progressive tax. The optimal tax itself need not be progressive, but it has a progressive upper bound and it coincides with the upper bound as soon as the latter is efficient.¹²

6. Conclusion

Should we help the low skilled or reward the hardworking? Our analysis is meant to shed light on the implications of this ethical dilemma.

First, recall that the approach adopted in this paper defines social preferences that refer only to individual noncomparable preferences and yet satisfy the Pareto principle and fairness requirements. Such social preferences can be applied in second-best contexts. Consequently, applications need not depend on delicate choices of individual utility functions, and the only data required are the joint distribution of wages and preferences in the population.

Our main result is that, in spite of a definite antagonism between these two goals, as illustrated in the incompatibility between the corresponding requirements in our model, there are interesting convergences between

¹²When, as in our finite model, W is not an interval, it still contains points of this convex function (for all values $w \in W$), so that this general shape is still relevant.

egalitarian-equivalent preferences (which focus on the low skilled) and equivalent-budget preferences (which focus on the hardworking), as far as the optimal tax is concerned. It is as if the second-best context reduced the intensity of this value conflict, as compared to the first-best context.

The tax schemes that turn out to be justified by our analysis have one of the following features: they maximize the basic income, they exhibit a zero marginal tax rate on incomes below the minimum wage (the minimum wage being defined as the earnings of agents working full time at the smallest wage rate) or they are below an upper bound exhibiting increasing marginal tax rates.

In the above theoretical analysis as well as in potential applications, the determination of the smallest wage rate plays an important role. The minimum legal wage seems a natural candidate (assuming law is respected), but one may also suggest that agents who are constrained in their labor time by an insufficient labor demand, such as the unemployed, may be treated as if their wage rate was zero above their current labor time. Unemployed agents are then equivalent to agents with a zero wage. As a consequence of our analysis, one might suggest that in times of high unemployment, it is reasonable to maximize the minimum income, whereas when unemployment is low it is advisable to seek low marginal rates of taxation for low incomes. A combination of unemployment allowances and tax exemption for low incomes could approximately achieve this goal without requiring any institutional adjustment during the business cycle.¹³

Appendix: Proofs

Proof of Proposition 1: Assume that the constraint $(y_{22}, c_{22}) R_{22}^*(y_{12}, c_{12})$ is not binding, so that $(y_{22}, c_{22}) P_{22}^*(y_{12}, c_{12})$. The social preferences under consideration are of the maximin type. Therefore, either equality among the relevant indices is achieved, or an incentive compatibility constraint is binding. Let U_2 be a utility function representing R_2 . As the above incentive constraint is not binding, one must have either

$$IT\left(\left(\frac{y_{12}}{w_1}, c_{12}\right), w_1, R_2\right) = IT\left(\left(\frac{y_{22}}{w_2}, c_{22}\right), w_1, R_2\right),$$

or

$$IT\left(\left(\frac{y_{12}}{w_1}, c_{12}\right), w_2, R_2\right) = IT\left(\left(\frac{y_{22}}{w_2}, c_{22}\right), w_2, R_2\right),$$

¹³Special observable characteristics of unemployed or handicapped agents may justify that they be given a special treatment, with a greater income support, even when these characteristics are not perfectly observed. See Salanié (2002).

or

$$U_2^{\max} \left(IB \left(\left(\frac{y_{12}}{w_1}, c_{12} \right), w_1, R_2 \right) \right) = U_2^{\max} \left(IB \left(\left(\frac{y_{22}}{w_2}, c_{22} \right), w_2, R_2 \right) \right),$$

implying $(\frac{y_{12}}{w_1}, c_{12}) I_2 (\frac{y_{22}}{w_2}, c_{22})$. As $w_1 < w_2$, this implies $(y_{12}, c_{12}) R_{22}^* (y_{22}, c_{22})$, in contradiction with the above assumption. ■

Proof of Proposition 2: By Proposition 1, $(y_{22}, c_{22}) I_{22}^* (y_{12}, c_{12})$ in both cases. By assumption, $(y_{12}, c_{12}) = (y_{21}, c_{21})$. That is, $w_1 - EE$ and $R_2 - EB$ can only differ with respect to the relative treatment of agents 11 and 12. But

$$\begin{aligned} IT \left(\left(\frac{y_{11}}{w_1}, c_{11} \right), w_1, R_1 \right) &\geq IT \left(\left(\frac{y_{12}}{w_1}, c_{12} \right), w_1, R_2 \right) \Leftrightarrow \\ IB \left(\left(\frac{y_{11}}{w_1}, c_{11} \right), w_1, R_1 \right) &\geq IB \left(\left(\frac{y_{12}}{w_1}, c_{12} \right), w_1, R_2 \right) \Leftrightarrow \\ U_2^{\max} \left(IB \left(\left(\frac{y_{11}}{w_1}, c_{11} \right), w_1, R_1 \right) \right) &\geq U_2^{\max} \left(IB \left(\left(\frac{y_{12}}{w_1}, c_{12} \right), w_1, R_2 \right) \right) \end{aligned}$$

for any utility function U_2 representing R_2 . Therefore, either equality of the indices is achieved in both cases, or the incentive constraint is binding in both cases. ■

Proof of Theorem 2: Consider a tax function τ^* defined by the fact that the income function $y - \tau^*(y)$ is continuous, nondecreasing, and the part of it lying in the positive orthant is the envelope curve of the population's indifference curves in (y, c) space, at an allocation z^* maximizing $\min_i IT(z_i, \tilde{w}, R_i)$. Let $y^* \geq 0$ be the smallest level of earnings such that $y - \tau^*(y) \geq 0$. By the No Identification assumption, over $[y^*, w_m]$ it is the envelope curve of agents from the w_m subpopulation. Let

$$\begin{aligned} IT_m &= \min \left\{ t \in \mathbb{R} \mid \exists \ell \in \left[\frac{y^*}{w_m}, 1 \right], t + \tilde{w}\ell = w_m\ell - \tau^*(w_m\ell) \right\} \\ &= \min \left\{ y - \tau^*(y) - \frac{\tilde{w}}{w_m}y \mid y \in [y^*, w_m] \right\}. \end{aligned}$$

IT_m is the minimum value of $IT(z_i, \tilde{w}, R_i)$ over the w_m subpopulation. Without loss of generality, we can assume that over $[0, y^*)$ the slope of $y - \tau^*(y) < 0$ is smaller than $\tilde{w}/w_m$, so that one simply has

$$IT_m = \min \left\{ y - \tau^*(y) - \frac{\tilde{w}}{w_m}y \mid y \in [0, w_m] \right\}.$$

Similarly, for agents with a higher w , the minimum value of $IT(z_i^*, \tilde{w}, R_i)$ is at least as great as

$$\min \left\{ y - \tau^*(y) - \frac{\tilde{w}}{w} y \mid y \in [0, w] \right\}.$$

Since $y - \tau^*(y)$ is nondecreasing, the above expression is nondecreasing in w , so that IT_m is indeed the minimum value of $IT(z_i^*, \tilde{w}, R_i)$ over the whole population.

Notice that, at a laissez-faire allocation,

$$\min \left\{ y - \frac{\tilde{w}}{w_m} y \mid y \in [0, w_m] \right\} = w_m - \tilde{w},$$

so that since z^* maximizes $\min_i IT(z_i, \tilde{w}, R_i)$, one has $IT_m \geq w_m - \tilde{w}$.

Let a new tax be defined by $\tau^{**}(y) = \max\{\tau^*(y), -IT_m - (\tilde{w} - w_m)\}$. This tax function is feasible, because it cuts all subsidies greater than a constant, $IT_m + (\tilde{w} - w_m)$, so that no agent's tax may decrease (and no subsidy increase), even after they adjust their choice. Moreover,

$$\begin{aligned} & \min \left\{ y - \tau^{**}(y) - \frac{\tilde{w}}{w_m} y \mid y \in [0, w_m] \right\} \\ &= \min \left\{ \begin{array}{l} \min \left\{ y - \tau^*(y) - \frac{\tilde{w}}{w_m} y \mid y \in [0, w_m] \right\}, \\ \min \left\{ y + IT_m + (\tilde{w} - w_m) - \frac{\tilde{w}}{w_m} y \mid y \in [0, w_m] \right\} \end{array} \right\} \\ &= \min \left\{ \begin{array}{l} \min \left\{ y - \tau^*(y) - \frac{\tilde{w}}{w_m} y \mid y \in [0, w_m] \right\}, \\ IT_m \end{array} \right\} \\ &= IT_m. \end{aligned}$$

Since the minimum value of $IT(z_i, \tilde{w}, R_i)$ over the w_m subpopulation, under application of τ^{**} , is at least as great as this expression, one obtains that τ^{**} maximizes $\min_i IT(z_i, \tilde{w}, R_i)$ as well.

By construction, for all $y \geq 0$, $\tau^{**}(y) \geq -IT_m - (\tilde{w} - w_m)$, and as shown above, for all $y \leq w_m$, $y - \tau^{**}(y) - \frac{\tilde{w}}{w_m} y \geq IT_m$, so that for all $y \leq w_m$,

$$-IT_m - (\tilde{w} - w_m) \leq \tau^{**}(y) \leq -IT_m + y - \frac{\tilde{w}}{w_m} y.$$

For $y = w_m$, one obtains $\tau^{**}(w_m) = -IT_m - (\tilde{w} - w_m)$, implying that for all $y \geq 0$, $\tau^{**}(y) \geq \tau^{**}(w_m)$. And for all $y \in [0, w_m]$,

$$\begin{aligned} \tau^{**}(y) &\leq y - \frac{\tilde{w}}{w_m} y - IT_m = y - \frac{\tilde{w}}{w_m} y + \tau^{**}(w_m) + \tilde{w} - w_m \\ &\leq \tau^{**}(w_m) + (w_m - y) \left(\frac{\tilde{w}}{w_m} - 1 \right). \end{aligned}$$

Finally, since $y - \tau^{**}(y)$ is nondecreasing, for all $y > w_m$ one has $\tau^{**}(y) \leq \tau^{**}(w_m) - w_m + y$.

It remains to note that since $IT_m = w_m - \tilde{w} - \tau^{**}(w_m)$, maximizing IT_m is equivalent to minimizing $\tau^{**}(w_m)$. ■

Proof of Theorem 3: Consider a tax function τ^* such that the income function $y - \tau^*(y)$ is continuous, nondecreasing, and the part of it lying in the positive orthant coincides with the envelope curve of the agents' indifference curves at z^* . By the No Identification assumption, the part of this envelope curve over some interval $[0, w]$, for $w \in W$, is also the envelope curve of the indifference curves of the subpopulation with a wage rate no greater than w .

Let

$$\tilde{U}_0 = \min_i \tilde{U}^{\max}(IB(z_i^*, w_i, R_i)).$$

Necessarily, $\tilde{U}_0 \geq \tilde{U}^{\max}(B^f(0, w_m))$, since a laissez-faire allocation z^{LF} is such that $\tilde{U}^{\max}(IB(z_i^{\text{LF}}, w_i, R_i)) = \tilde{U}^{\max}(B^f(0, w_m)) \geq \tilde{U}(0, 0)$ for all i . An allocation at which $\min_i \tilde{U}^{\max}(IB(z_i, w_i, R_i))$ is maximized cannot do worse. Notice that

$$\tilde{U}_0 = \tilde{U}^{\max}(B^f(\tilde{E}(w_m, \tilde{U}_0), w_m)) \geq \tilde{U}^{\max}(B^f(0, w_m))$$

implies $\tilde{E}(w_m, \tilde{U}_0) \geq 0$.

Take any $w \in W$, and any $y \leq w$. By the above observation about the envelope curve, there is an agent i such that $w_i \leq w$, and such that $z_i^* I_i(\frac{y}{w_i}, y - \tau^*(y))$, which entails

$$IB(z_i^*, w_i, R_i) \subseteq B^f(-\tau^*(y), w_m).$$

Since $\tilde{U}^{\max}(IB(z_i^*, w_i, R_i)) \geq \tilde{U}_0$,

$$B^f(-\tau^*(y), w_i) \supseteq B^f(\tilde{E}(w_i, \tilde{U}_0), w_i),$$

or equivalently, $-\tau^*(y) \geq \tilde{E}(w_i, \tilde{U}_0)$. As $\tilde{E}(w_i, \tilde{U}_0)$ is nonincreasing in w_i , and $w \geq w_i$, one obtains $-\tau^*(y) \geq \tilde{E}(w, \tilde{U}_0)$. Take any $y > w$. Since $y - \tau^*(y)$ is nondecreasing, one has

$$y - \tau^*(y) \geq w - \tau^*(w) \geq w + \tilde{E}(w, \tilde{U}_0),$$

and therefore $\tau^*(y) \leq y - w - \tilde{E}(w, \tilde{U}_0)$. This achieves the proof that for all $y \geq 0$, all $w \in W$,

$$\tau^*(y) \leq (y - w)^+ - \tilde{E}(w, \tilde{U}_0).$$

Define a new tax function τ^{**} by $\tau^{**}(y) = \max\{\tau^*(y), -\tilde{E}(w_m, \tilde{U}_0)\}$. Since it cuts all subsidies greater than $\tilde{E}(w_m, \tilde{U}_0)$, this new tax function is feasible. In addition, it also satisfies the property that for all $y \geq 0$, all $w \in W$,

$$\tau^{**}(y) \leq (y - w)^+ - \tilde{E}(w, \tilde{U}_0).$$

This is due to the fact that if $\tau^{**}(y) = \tau^*(y)$, this immediately follows from the same property of τ^* ; and if $\tau^{**}(y) = -\tilde{E}(w_m, \tilde{U}_0)$, then the conclusion is obtained by observing that

$$\tilde{E}(w, \tilde{U}_0) - \tilde{E}(w_m, \tilde{U}_0) \leq 0 \leq (y - w)^+.$$

We now show that at any (incentive-compatible) allocation z^{**} generated by τ^{**} , one has

$$\min_i \tilde{U}^{\max}(IB(z_i^{**}, w_i, R_i)) = \tilde{U}_0.$$

First, we note that for all $y \leq w_m$, $\tau^{**}(y) = \tau^*(y) = -\tilde{E}(w_m, \tilde{U}_0)$, since $\tau^*(y) \leq -\tilde{E}(w_m, \tilde{U}_0)$. This entails that all agents from the w_m subpopulation have the implicit budget $B^f(\tilde{E}(w_m, \tilde{U}_0), w_m)$. For all those agents,

$$\tilde{U}^{\max}(IB(z_i^{**}, w_i, R_i)) = \tilde{U}^{\max}(B^f(\tilde{E}(w_m, \tilde{U}_0), w_m)) = \tilde{U}_0.$$

More generally, since $(\tau^{**})'(y) = 0$ over $[0, w_m]$ and individual preferences are convex, one can see that for all agents i with $y_i^{**} \leq w_m$, z_i^{**} is the best for R_i over $B^f(\tilde{E}(w_m, \tilde{U}_0), w_i)$, so that

$$IB(z_i^{**}, w_i, R_i) = B^f(\tilde{E}(w_m, \tilde{U}_0), w_i),$$

and as $w_m \leq w_i$, $\tilde{E}(w_m, \tilde{U}_0) \geq \tilde{E}(w_i, \tilde{U}_0)$, implying

$$IB(z_i^{**}, w_i, R_i) \supseteq B^f(\tilde{E}(w_i, \tilde{U}_0), w_i),$$

and then

$$\tilde{U}^{\max}(IB(z_i^{**}, w_i, R_i)) \geq \tilde{U}^{\max}(B^f(\tilde{E}(w_i, \tilde{U}_0), w_i)) = \tilde{U}_0.$$

Now consider an agent i with $y_i^{**} > w_m$. Then $w_i > w_m$. One has, for all $y \leq w_i$,

$$z_i^{**} R_i \left(\frac{y}{w_i}, y - \tau^{**}(y) \right).$$

If for all $y \leq w_i$, $\tau^*(y) \leq -\tilde{E}(w_m, \tilde{U}_0)$, then for all $y \leq w_i$, $\tau^{**}(y) = -\tilde{E}(w_m, \tilde{U}_0)$ and

$$z_i^{**} R_i \left(\frac{y}{w_i}, y + \tilde{E}(w_m, \tilde{U}_0) \right),$$

so that

$$IB(z_i^{**}, w_i, R_i) \supseteq B^f(\tilde{E}(w_m, \tilde{U}_0), w_m),$$

and, since

$$B^f(\tilde{E}(w_m, \tilde{U}_0), w_i) \supseteq B^f(\tilde{E}(w_m, \tilde{U}_0), w_m),$$

one obtains $\tilde{U}^{\max}(IB(z_i^{**}, w_i, R_i)) \geq \tilde{U}_0$.

If there exists $y \leq w_i$ such that $\tau^*(y) > -\tilde{E}(w_m, \tilde{U}_0)$, then, for any such y , $\tau^{**}(y) = \tau^*(y)$. Let

$$y_0 = \arg \max_{0 \leq y \leq w_i} \tau^*(y).$$

Necessarily, $IT(z_i^{**}, w_i, R_i) \geq -\tau^*(y_0)$, because

$$IT(z_i^{**}, w_i, R_i) \geq -\max_{0 \leq y \leq w_i} \tau^{**}(y) = -\tau^*(y_0).$$

Recollect that $y - \tau^*(y)$ is the envelope curve of the agents' indifference curves at z , and that, by the No Identification assumption, over the interval $[0, w_i]$, it is the envelope curve for the subpopulation with wage no greater than w_i . Then there exists an agent j with wage $w_j \in [y_0, w_i]$, such that

$$IT(z_j, w_j, R_j) \leq -\tau^*(y_0) \leq IT(z_i^{**}, w_i, R_i),$$

implying

$$IB(z_i^{**}, w_i, R_i) \supseteq IB(z_j, w_j, R_j).$$

Since one has $\tilde{U}^{\max}(IB(z_j, w_j, R_j)) \geq \tilde{U}_0$, this implies $\tilde{U}^{\max}(IB(z_i^{**}, w_i, R_i)) \geq \tilde{U}_0$.

The tax function τ^{**} satisfies the constraint enunciated in the theorem. It remains to show that one must seek to minimize $\tau^{**}(w_m)$. This is an immediate consequence of the fact that

$$\min_i \tilde{U}^{\max}(IB(z_i^*, w_i, R_i)) = \tilde{U}_0 = \tilde{U}^{\max}(B^f(-\tau^{**}(w_m), w_m)).$$

■

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